

Exploratory Data Analysis on the Automobile Dataset Report

Introduction

The Automobile Dataset contains 205 rows and 26 columns of car specifications including price, engine size, fuel efficiency and body style. The goal was to clean the data, handle missing values, answer key questions about expensive cars, fuel efficiency, engine capacity and lastly manufacturer model counts.

Data cleaning

- Removed two columns which is “normalized-losses” and “symboling”
- No duplicate rows were found
- Replaced “?” with NaN and dropped any row with missing values.
- Converted numeric columns which are price, engine size, city mpg etc into int64.

Missing data

Columns with ‘?’ values included:

- normalized-losses(41)
- price(4)
- stroke(4)
- horsepower(2)
- peak-rpm(2)
- num-of-doors(2)

After the removal of rows with any ‘?’ values we had the final clean dataset of **193 rows**.

Data stories and visualisations

Five most expensive cars

1. Mercedes-benz hardtop which goes for \$45400
2. Bmw sedan which goes for \$41315
3. Mercedes-benz sedan which goes for \$40960
4. Porsche convertible which goes for \$37028
5. Bmw sedan which goes for \$36880

We see that the most expensive car has the least engine size of 304 than the second most expensive car which is the sedan has 308 engine size. This suggests the brand and body style has an impact on price more than the engine size alone.

Most fuel efficient manufacturer

Chevrolet had the highest average combined MPG where city mpg was added with highway mpg divided by 2. The Bar plot shows Chevrolet with the highest bar, then followed by Honda and Volkswagen. We also see that the bar plot shows a steep drop after the top three, which reveals that most manufacturers prioritize performance/size over fuel economy.

Largest engine capacity

There are top five vehicles with the largest engine size which are:

1. Jaguar sedan engine size of 326 highest horsepower of 262

2. Mercedes-benz sedan engine size of 308 horsepower of 184
3. Mercedes-benz hardtop engine size of 304 horsepower of 184
4. Jaguar sedan engine size of 258 horsepower of 176
5. Jaguar sedan engine size of 258 horsepower of 176

The interesting thing that stood out is that Jaguar has the highest engine capacity and horsepower however is not the most expensive car in the dataset. The Mercedes-Benz hardtop is the more expensive with a low engine capacity. This shows that luxury branding and body styles can command higher prices even with smaller engines.

Manufacturer with most models

Toyota has the most models , 32. It is followed by Nissan which has 18 models. This also tells us that Toyota offers a variety of body styles, engine options and price points to appeal to diverse customers. Honda shares the same models as Mitsubishi of 13 models.

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